TONE DISCRIMINATION IN WHITE RATS

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The purpose of this article is to present evidence showing that it is possible for white rats to form discrimination habits with tones as stimuli.

The animals in Hunter's elaborate and carefully checked experiments (3, 4, 5, 6) failed in all cases but one to form such habits when tones were used as stimuli, while they succeeded when noises were used as stimuli. Hunter used a T-shaped discrimination box; he produced the tones by blowing whistles and striking tuning forks. All of his stimuli were diffuse, that is, there was no element of direction in them, a circumstance which he considers favorable (4, p. 317). (See also Miss Barber's experiments (1) in which localization of sound stimuli was a factor; her results are in agreement with Hunter's.) His animals were trained to turn to one side of the discrimination box when an auditory stimulus was given and to turn to the other side when there was silence. The noises were all found to be effective in establishing the habit. So was the whistle, although a checking up indicated that the noise of the rushing air rather than the tone was the effective element in this case. The tuning fork was ineffective in producing the habit in all animals but one. Hunter concluded that "the great mass of negative data make it certain that the rat's sensitivity to tonal stimuli between the vibration frequencies of 256 d.v. to 4096 d.v. is very slight, if it does exist" (6, p. 185). It should be noted especially that his evidence is based upon discontinuous tones produced by striking tuning forks; electrically driven tuning forks were not used by him, although he refers to this possibility.

Our experiments differed from Hunter's chiefly in the use of a continuous tone, diffuse in one situation and with a directive

factor in another, and in the use of a Y-shaped discrimination box. The results were positive.

APPARATUS AND METHODS

The tone producing apparatus consisted of a Hartley oscillating circuit plus a two-stage amplifier. In the diffuse tone situation a single balanced armature loudspeaker was mounted separately from the discrimination box, about 1.5 m. behind the forking of the runway and 1 m. above it. In the directive tone situation two identical magnetic diaphragm loudspeakers were placed into the box, imbedded in cotton, where the lights would be if one were to use the box for visual discrimination.

Noises were used only in the diffuse tone situation. They were produced by a buzzer and a whistle that were mounted in the place of the loudspeaker.

The tone produced by the oscillator was kept at a frequency of about 1000 d.b., except where otherwise noted. The whistle had a frequency of 2100 d.v. Its tone is called a noise here on account of Hunter's results with whistles. The amplifier was always set at maximum intensity (199 tubes, B battery 22.5 volts).

The main source of errors with the tone lay in the capacity of the oscillator to pick up signals from the make and break in the primary circuit of the induction coil which provided the electric shock used for punishment. There were three ways to overcome this difficulty: (*a*) The use of a properly transformed lighting current (2a) in place of the induction coil; (*b*) shielding and grounding the oscillator, and (*c*) turning on the shock only after the animal had chosen the wrong alley. The last, being the safest, was used in the chief experiment (II).

When these three stimuli (from the oscillator, the whistle, and the buzzer) were compared by means of an oscillograph it was found that the picture of the buzzer sound was as irregular as expected. The line of the oscillator tone was very clean cut with regular oscillations. The tone of the whistle showed the same regular oscillations, but its line did not appear quite so clean cut as that of the oscillator tone. Whether or not this was due to air noises was not determined.

The discrimination box was Y-shaped as shown in figure 1. During the experiments with the diffuse stimuli the box was uncovered. During the experiments with the directive tone (two loudspeakers) it was completely covered except for the starting compartment up to the forking of the alleys. In this situation the alley entrances were partly closed, leaving openings 4 inches wide and 2 inches high. Sound reflectors consisting of boards sloping upwards at an angle of 45° and inwards towards the loudspeakers were attached to the upper edges of the openings.

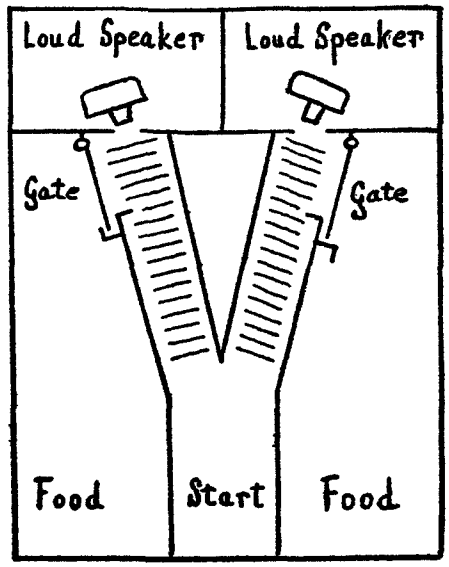


FIG. 1. DIAGRAM OF THE Y-SHAPED TONE DISCRIMINATION BOX

A wire screen was used as a stop gate in the wrong alley. In a later part of the experiment another stop gate was placed into the right alley and left unlocked. These gates could not be seen from the place where the animal had to make its choice.

Twenty animals were used. All were four to five weeks old when the trials started.

The usual method of irregular alternations of right and left turns combined with reward and punishment was used.

The experimenter always placed the animal into the starting

compartment of the discrimination box with the left hand so that the animal's head pointed towards the right wall.

The reward consisted of pieces of cheese placed in both food compartments. The animals were permitted to feed for thirty seconds in each trial.

The punishment consisted of the usual electric shock produced by an induction coil, except where otherwise noted. In the first experiment the shock was turned on all the time. In the second experiment it was turned on only after the animal had made a wrong choice.

For the sake of comparability with Hunter's results our criterion for the presence of discrimination was the same as in Hunter's later experiments, namely an average of 90 per cent of correct turns for three consecutive series of 10 trials with no series below 80 per cent. However, it appears to us that a less severe criterion, say 80 per cent correct turns in three consecutive series, would be justifiable in view of the difficulty of the habit. We sometimes refer to the latter as the easier criterion.

EXPERIMENT I. DISCRIMINATION WITH A DIFFUSE TONE

a. In an exploratory experiment we duplicated Hunter's conditions in most respects. Two rats (nos. 1, 2) were trained to go to the right when a buzzer was sounded and to the left when there was silence. After the criterion had been attained the oscillator tone was sounded simultaneously with the buzzer and finally the buzzer was eliminated. When the tone was then sounded alone both rats attained the criterion in their first thirty trials.

In order to check these results the following modifications were introduced: (*a*) A screen was placed between the experimenter and the box in order to eliminate possible cues from the experimenter; result negative. (*b*) The shock was stopped entirely in order to eliminate possible impurities of the tone; result negative. (Quantitative results with similar check experiments will be reported only with the second experiment.)

b. On the basis of this satisfactory exploration we decided to train some rats with the radio tone without first training them

with the buzzer. Together with them we decided also to train some rats with the buzzer and the whistle as a control, giving them the same daily treatment as the tone group. The whistle was considered as a control on account of Hunter's positive results with it. In the tone group we had five rats (nos. 3 to 7) and in the control group we had four, one to be trained with the buzzer (no. 8) and three with the whistle (nos. 9 to 11).

All animals were trained to go to the right when the auditory stimulus was sounded, and to the left when there was silence.

The criterion of learning was attained by the first four animals of the tone group in the following trials: 170, 170, 190, 180. The last one of the group was given 320 trials of training without success as far as Hunter's criterion was concerned; however, it attained the easier criterion (80 per cent correct trials in 3 successive series of 10 trials) in 150 trials; it also mastered the situation in experiment II.

The control group learned the habit as follows: The rat trained with the buzzer in 140 trials; and the rats trained with the whistle in 190, 130, and 180 trials respectively.

Since there is no significant difference between the two groups we may assume that the tone used is equivalent to a noise in the acquisition of an auditory discrimination habit. (Cf. Upton (8) on this point.)

The control group was then given the tone in place of the buzzer and whistle with the result that the rats trained originally with the whistle all attained Hunter's criterion and that the rat trained with the buzzer only attained the easier criterion.

In order to check on the presence of other cues the following modifications were introduced: Screen between box and experimenter, substitution of properly transformed lighting current for the induction coil circuit, removal of the stop gate in the wrong alley, and very irregular series. The result was negative in all cases. (Quantitative results with check experiments will be given below.)

EXPERIMENT II. DISCRIMINATION WITH A DIRECTIVE TONE

This experiment was performed like a black-white discrimination experiment, except that two tone stimuli were used instead

of two light stimuli. By means of identical loudspeakers in both alleys (fig. 1) and a double throw switch it was possible to have one alley tone-filled and the other comparatively silent. This we have called the directive tone situation to distinguish it from the diffuse tone situation in the previous experiment where the tone was equally loud in both alleys whenever it was sounded. In the present situation with the directive tone the animal had to choose in every trial between the tone-filled alley and the silent one; some animals were trained to avoid the tone and others to go towards it.

In order to equalize conditions for all animals the series of turns used during the training period was R(ight) L(eft) LRRLRRLL on odd numbered days, and LRRLLRLLRR on even numbered days.

The shock was not turned on until the animal had chosen and entered the wrong alley.

a. In the first part of the experiment the old animals (nos. 2, 3, 4, 5, 7, 8, 9, 11) were used that had already been trained in the diffuse tone situation. They attained Hunter's criterion as follows:

Rats 2, 3, 4, 5, 7, 8 and 9 in 220, 90, 220, 180, 210, 230 and 60 trials respectively.

Rat 11 was given 230 trials and failed to reach this criterion as well as the easier one.

Animals 1, 4, 5, and 8 were trained to avoid the tone, the others to go towards it. No significant difference was found between the two groups.

b. In the second part of the experiment 9 untrained young rats were used, nos. 12 to 20. Five of these, nos. 12 to 16, were trained to avoid the tone and the rest, nos. 17 to 20, to go towards it. The animals attained Hunter's criterion as follows:

Rats 13, 14, 15, 16, 17, 18, 19 and 20 in 250, 220, 230, 140, 100, 80, 110 and 140 trials respectively.

Rat 12 never attained Hunter's criterion; it reached the easier one in 320 trials and never dropped below 80 per cent until the end of its training at 390 trials. It might therefore also be considered as having learned the discrimination.

The difference between the two sub-groups (away from, and towards the tone) is probably not due to the difference in the procedure, but rather to the difference in the litters. Nos. 17 to 20 came from one litter and nos. 12 to 16 came from two other litters. We are prompted to hazard this suggestion, because we were impressed during the entire experiment with such individual differences between litters. Since this habit seems to be just below the limit of the rat's learning ability, individual differences were brought out rather sharply.

Since it was observed that the animal's correctness of response varied considerably even after the criterion of learning had been reached, it was decided to continue the training in some cases in order to see whether there appeared any approximation to the kind of automatization observed usually in the black-white discrimination habit. The training of rats 17 and 18 was continued up to 690 and 680 trials respectively. During their last 200 trials the animals made the following number of errors per fifty trials: 2, 3, 2, 6, and 4, 5, 7, 4 respectively. Complete automatization was never observed.

c. The following check experiments were made in order to find out whether or not the animals responded to any cues other than the tone.

1. In order to eliminate cues from the experimenter, an assistant took the animals from the food compartment and placed them in the starting compartment after each trial and then stepped aside about 6 feet so that he could not see the rat nor be seen by him, while the experimenter made his observations from an adjoining room during the entire series. The results were as follows:

Rat 17 made 9, 8, correct responses per series.

Rat 18 made 9, 10, correct responses per series.

Rat 19 made 7, 9, 8, 10 correct responses per series.

Rat 20 made 8, 10, 8, 10 correct responses per series.

These results indicate complete indifference of the animals to cues from the experimenter. They stand in contrast to Johnson's results with cats and dogs (7).

A less severe check than this one consisted in the use of a

screen with an aperture, placed between the experimenter and the box. We did not notice any change in such a case.

2. Since we used two alternating training series, it seemed possible that the animal might have learned at least part of the series. In order to check this possibility four highly irregular series were used, the same that Hunter used for a similar purpose, namely: LRLLRRRLLR, RLLLRRLRRL, LLRRLLRRLR, and RLLRRRLLLR. The results were as follows:

Rat 16 made 9, 8, 9, 9 correct responses per series.

Rat 17 made 10, 10, 10, 9 correct responses per series.

Rat 18 made 7, 9, 9, 9 correct responses per series.

Rat 19 made 8, 8, 10, 6 correct responses per series.

Rat 20 made 8, 10, 8, 10 correct responses per series.

These results show that the animals' responses did not depend upon subtle position habits.

3. Although the stop gate in the wrong alley was not visible from the place where the animal had to make his choice, it seemed possible that he might derive an auditory cue from it (e.g., his breathing sounding differently in the stopped alley than in the open one). The gate was therefore removed. In this situation rat 17 made 9, 8, 9 correct responses, and rat 18 made 8, 8, 9 correct responses per series. Since this method was rather inconvenient and introduced another factor, namely having food only in one compartment, the rest of the experiment was carried on with two identical stop gates, one in each alley. The one in the wrong alley was locked, while the one in the correct alley could be pushed open by the animal. Since this new situation of confronting a door in the correct alley and having to push it open disturbed the animals considerably at first (nos. 17, 18, 19, and 20 made only 6, 5, 4, and 5 correct responses respectively on the first day), the training was continued until all four animals had attained Hunter's criterion again. The result then was that the animals were able to respond correctly with practically identical conditions as far as the stop gates were concerned.

4. A commercial oscillator (Western Electric, 8–A) with perfect shielding and grounding was substituted for the one made by us in order to provide the best possible purity of tone in case our

apparatus should be faulty. During the first three days with tones of 1000, 2000, and 4000 d.v. per second the results were as follows:

Rat 17 made 9, 9, 9 correct responses per series.

Rat 18 made 8, 9, 9 correct responses per series.

Rat 19 made 8, 10, 7 correct responses per series.

Rat 20 made 9, 9, 10 correct responses per series.

Evidently the use of a better oscillator did not affect the animals adversely. Incidentally, this experiment extended the range of the tone.

5. Although the loudspeakers were imbedded in cotton and rested on the same floor it might have been possible, though highly improbable, for vibrations to be felt more in one alley than in the other, if vibrations were felt at all. During a whole section of the experiment the loudspeakers were therefore suspended from a frame which was independent of the discrimination box. No change in the animals' behavior was noticed; during the first day of this new situation each animal made 10 correct responses.

6. In order to find out directly whether the tone was the only cue to which the animals responded, the tone was omitted entirely one day with this result: Rats 17, 18, 19, and 20 made 5, 4, 3, and 4 correct responses respectively in 10 trials. There was no doubt in this case that the animals had been deprived of the significant factor in the stimulus situation.

SUMMARY OF RESULTS AND DISCUSSION

Eleven rats were trained in a Y-shaped discrimination box to turn to the right when a diffuse tone was sounded and to the left when there was silence; 9 of these attained the criterion of learning. Eight of these and 9 new ones were trained in the same box but with a situation requiring a discrimination between a tone-filled alley and one comparatively silent; 7 out of the first group and 8 out of the second attained the criterion of learning.

That these animals really did respond to the tone and not to other cues was then shown in a number of check experiments.

On the basis of this evidence we maintain that rats are capable of forming discrimination habits with tones as stimuli.

To explain the discrepancies between Hunter's results and ours is a difficult task. The possibilities are: (*a*) The Y-shaped box used by us as compared with the T-shaped box used by Hunter may give the animal a better chance in the directive tone situation to choose between the alternatives. (See below for a discussion of the behavior of choosing.) However, this would leave the positive results with the diffuse tone situation unexplained. (*b*) On account of its continuous character the oscillator tone employed by us may be superior to the intermittent striking of a tuning fork as was done by Hunter. The force of this argument is weakened, however, by the fact that Hunter showed that discontinuous as well as continuous noises are effective in producing discrimination habits. So why should there be a difference between continuous and discontinuous tones? We admit our inability to explain the discrepancies. But we should like to point out that Hunter's results disagree, while ours agree, with those obtained by the method of conditioning responses with other lower mammals (8, 9).

Finally, we should like to describe briefly the quality of the rats' behavior when choosing between the two pathways ("choosing" to be taken in an objective sense). Besides being another bit of evidence for the fact that the animals were guided by the tone, this behavior has a significance of its own. Usually, especially in the earlier trials, the rat would sniff before turning into an alley. Then the sniffing movements would stop and the rat would remain perfectly quiet for some time, often with the head turned so that one ear was oriented towards the back of the alley. The behavior was that of a "listening" animal. A most significant type of behavior might then occur: The rat would move its head or whole body from alley to alley, and in addition there might be a jerk towards the correct alley and a jerk away from the wrong alley, before it would enter one of the alleys. This type of behavior appeared to us to be a vestige of actual trial and error behavior, a first level perhaps of gestural thinking. It was studied in detail by one of the authors (2) who found that the frequency of its occurrence was in a definite relation to success and failure in the acquisition of the habit.

CONCLUSION

Evidence has been presented to show that white rats are capable of acquiring discrimination habits with tones as stimuli. They learn to discriminate between silence and a tone when these are presented successively, and also between a tone-filled alley and a silent one when these are presented simultaneously.

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